Indian Institute of Technology, Kanpur

Department of SPASE Laboratory

Report Of Radio Astronomy Laboratory: SPA617

ANTENNA CHARACTERIZATION

by

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Programme: PhD

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1. Aim

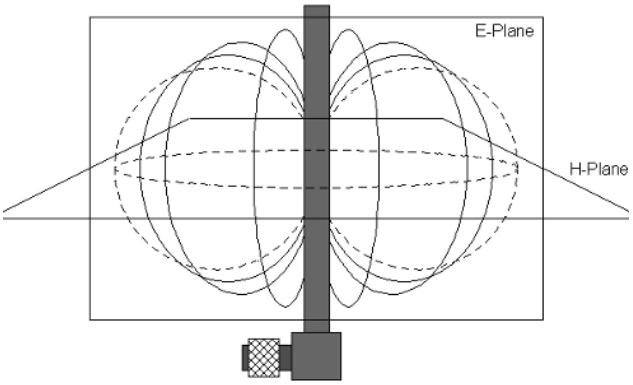
To characterize the given antenna by measuring its VSWR/return loss, received power variation with distance, and radiation beam pattern in co-polarization and cross-polarization configurations. The half-power beamwidth (HPBW) and an estimate of antenna directivity are also determined.

2. Apparatus Required

- Monopole and dipole antennas
- RF signal generator
- Digital storage oscilloscope (DSO)
- Spectrum analyser
- RF Tee connector
- 50 Ω character impedance termination
- Antenna support stand
- Coaxial cables and suitable RF adapters

3. Theory

An antenna is a transducer that converts electrical power into electromagnetic waves and vice versa. The radiation pattern represents how the radiated or received power varies with direction. It consists of a main lobe and may also contain side and back lobes. A dipole antenna has two equal metal arms in a straight line and is fed at the centre. Its total length is approximately half a wavelength (λ/2), with a feed line connected between the two arms and has very little radiation along the wire. A monopole antenna is approximately half of a dipole mounted above a conducting ground plane. Its length is approximately one-quarter wavelength (λ/4).The E-plane contains the electric-field and propagation vectors, while the H-plane contains the magnetic-field and propagation vectors.



For VSWR measurements, the measured voltage standing-wave ratio is

VSWR = Vmax / Vmin

The magnitude of the reflection coefficient is

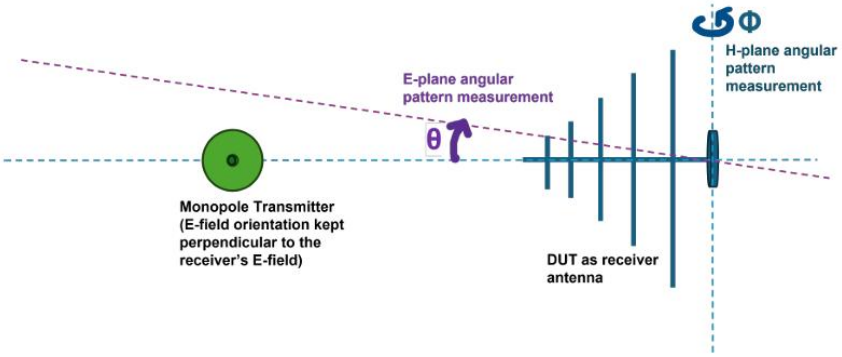
$|\Gamma| = \frac{(\text{Vmax} - \text{Vmin})}{(\text{Vmax} + \text{Vmin})}$ and the return loss is $RL = -20log_{10}|\Gamma|dB$

The half-power beamwidth (HPBW) is the angular width of the main lobe between the two −3 dB points, i.e. the points where the received power is half of the maximum power.

Directivity is a measure of how concentrated the radiation is in the direction of the main lobe compared with an isotropic antenna. The directivity is related to the beam solid angle ΩA as $D=\frac{4\pi}{\Omega A}$.

The directivity is estimated from the measured E- (θ_E) and H-plane (θ_H) HPBWs using

$D \approx \frac{4\pi}{(\theta E\ \theta H),}$ where θE and θH are in radians.



Beam pattern characterization set-up

4. Procedure

Part A: Measurement of VSWR, Return Loss and Operating Frequency

1. Connect the **signal generator, RF Tee connector, antenna (DUT), and oscilloscope** according to the VSWR measurement setup.
2. Set the oscilloscope input impedance to **1 MΩ**.
3. Keep the signal-generator amplitude constant and select the required test frequencies over the expected operating range of the antenna.
4. At each selected frequency, perform a small frequency sweep containing one complete amplitude-ripple period.
5. From the oscilloscope, record:
    - $V_{\max}$— maximum voltage observed.
    - $V_{\min}$— minimum voltage observed.
6. Use the **same voltage measurement type** for both $V_{\max}$and $V_{\min}$, such as peak-to-peak voltage.
7. Calculate the VSWR using: $VSWR = \frac{V_{\max}}{V_{\min}}$
8. Calculate the magnitude of the reflection coefficient:

$$|\Gamma| = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}$$

9. Calculate the return loss: $RL = -20\log_{10} | \Gamma |$

10. Repeat the measurements for the different selected frequencies.
11. Plot VSWR versus frequency and return loss versus frequency.
12. Identify the frequency corresponding to the minimum VSWR and take it as the approximate operating/resonant frequency.
13. Determine the frequency range over which:

$$VSWR \leq 2$$

Part B: Effect of Distance on Received Power

1. Connect the transmitting and receiving antennas according to the experimental setup.
2. Set a suitable operating frequency and signal-generator amplitude.
3. Adjust the length of the monopole according to the selected frequency.
4. Keep the transmitting and receiving antennas at the same height.
5. Align the antennas so that they have the same polarization.
6. Vary the separation distance between the transmitting and receiving antennas.
7. At each distance, record the received/peak power from the spectrum analyser.
8. Repeat the measurement for all the selected distances.
9. Plot: P_r versus R
   where P_r is received power and R is the separation distance.
10. Observe how the received power changes as the distance between the antennas increases.
11. Estimate the reduction in received power when the distance is doubled.

Part C: Cross-polarization Radiation/Beam Pattern

1. Keep the same experimental arrangement used for the co-polarization measurement.
2. Rotate/adjust the transmitting antenna so that the electric fields of the transmitting and receiving antennas are perpendicular to each other.
3. Maintain the same:
   - Frequency
   - Antenna separation
   - Antenna heights
   - Angular intervals
   - Instrument settings
4. Vary the E-plane angle θ.

5. Record the received power at each angular position.
6. Return the antenna to the reference position.
7. Vary the H-plane angle ϕ.
8. Record the received power for each angular position.
9. Plot the cross-polarized E-plane and H-plane radiation patterns as polar plots.
10. Compare the cross-polarization patterns with the corresponding **co-polarization** patterns.

Part D: Half-Power Beamwidth and Directivity

1. Use the **co-polarized E-plane radiation pattern** obtained in Part C.
2. Identify the maximum received power of the main lobe.
3. Determine the power level that is **3 dB below the maximum power**: $P_{-3dB} = P_{\max} - 3\ dB$
4. Locate the two angular positions at which the E-plane radiation pattern crosses this −3 dB level.
5. Calculate the E-plane half-power beamwidth: $\boxed{\theta_E = \theta_2 - \theta_1}$
6. Repeat the same procedure using the co-polarized H-plane radiation pattern.
7. Locate the two −3 dB points of the H-plane main lobe.
8. Calculate the H-plane half-power beamwidth: $\boxed{\theta_H = \theta_2 - \theta_1}$
9. Convert both beamwidths from degrees to radians: $\theta_{\text{rad}} = \theta_{\text{deg}} \frac{\pi}{180}$
10. Estimate the antenna directivity using: $\boxed{D \approx \frac{4\pi}{\theta_E \theta_H}}$

where θ_E and θ_H must be expressed in radians.

11. Compare the obtained beamwidths and directivity with the observed shapes of the radiation patterns

5. Observations

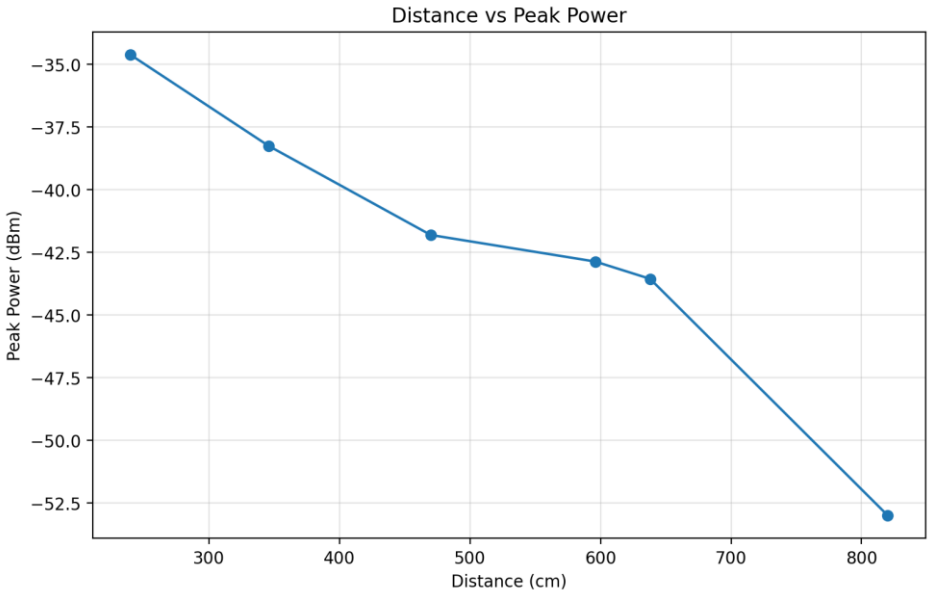
VSWR observations

Frequency	Vmax	Vmin	VSWR	|Γ|	Return loss (dB)	Remark
300 Mhz	3.00 V	2.00 V	1.50	0.200	13.98	Lower VSWR of the two readings
120 Mhz	4.50 V	0.40 V	11.25	0.837	1.55	Higher mismatch

Distance vs peak-power observations

Distance (cm)	Peak Power (dBm)
820	-53.00
638	-43.56
596	-42.87
470	-41.81
346	-38.26
240	-34.62

Figure 1. Distance vs peak power



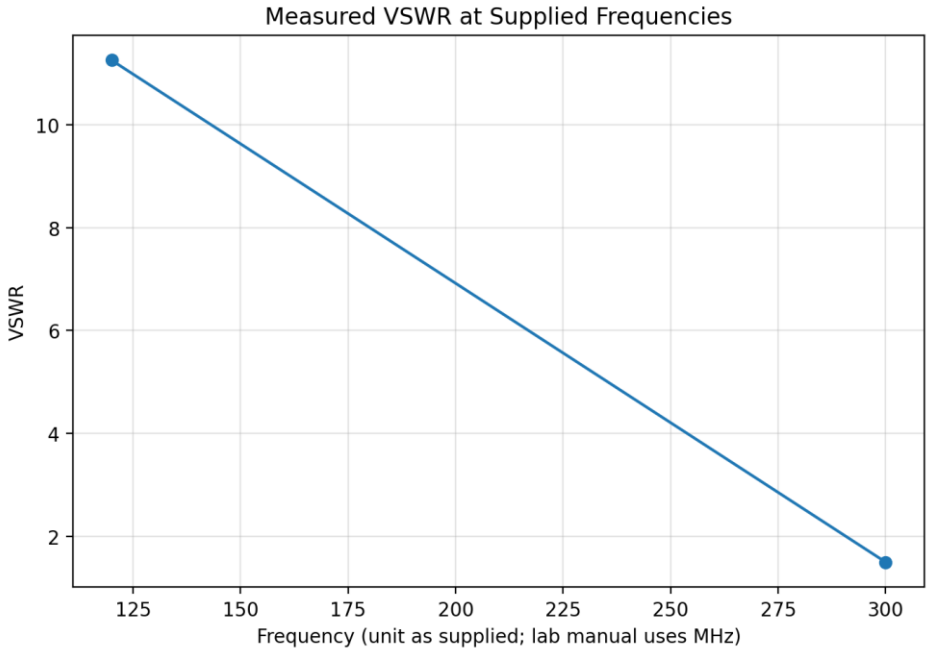
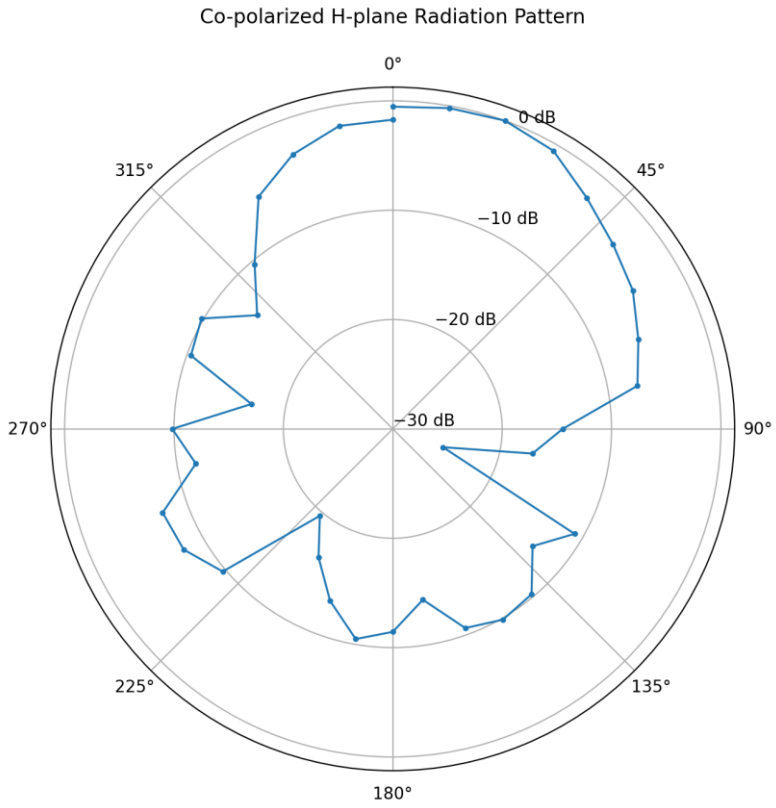
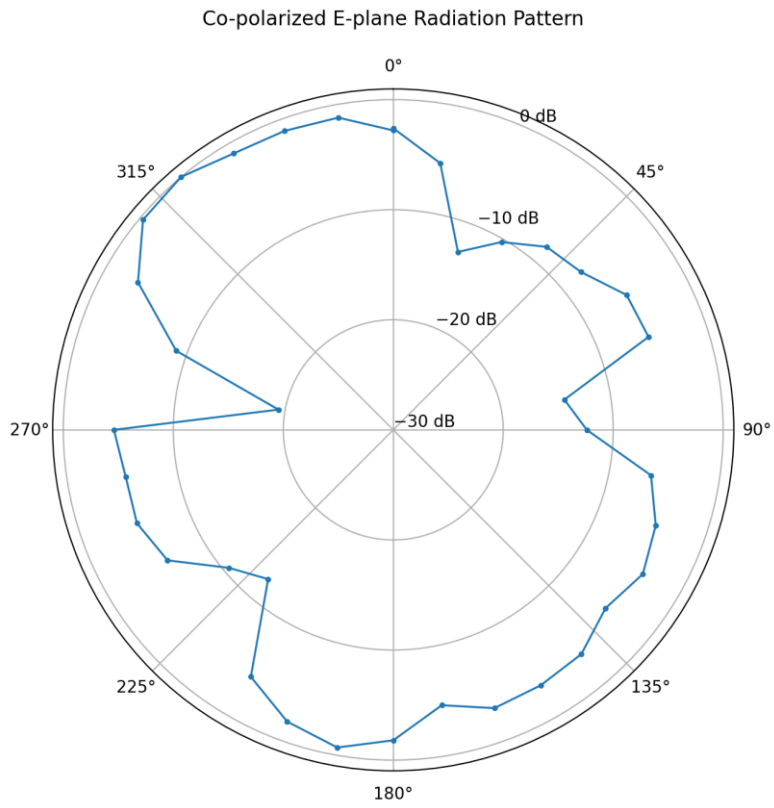


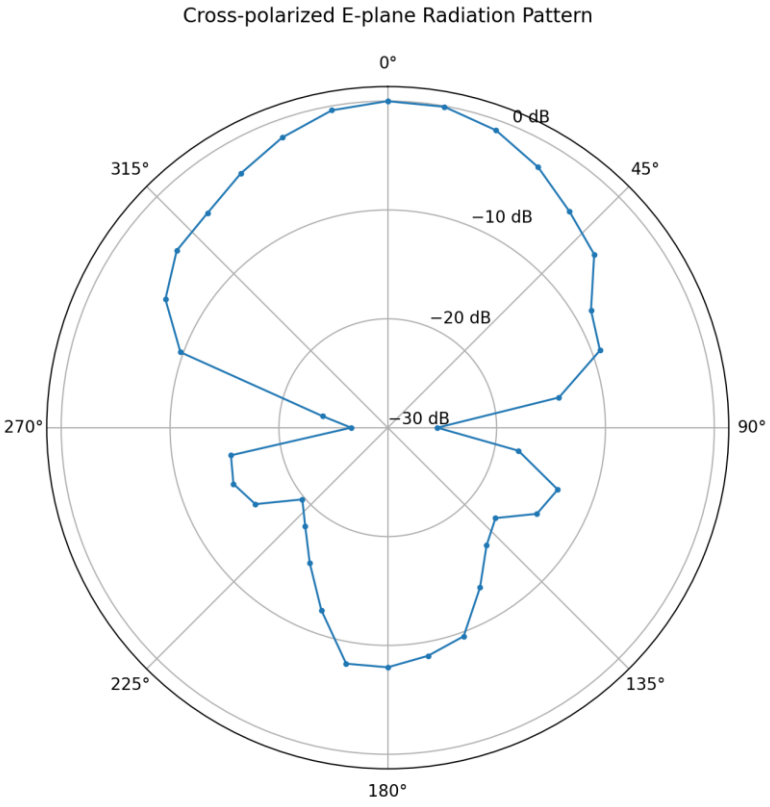
Figure 2. VSWR at the two supplied frequencies

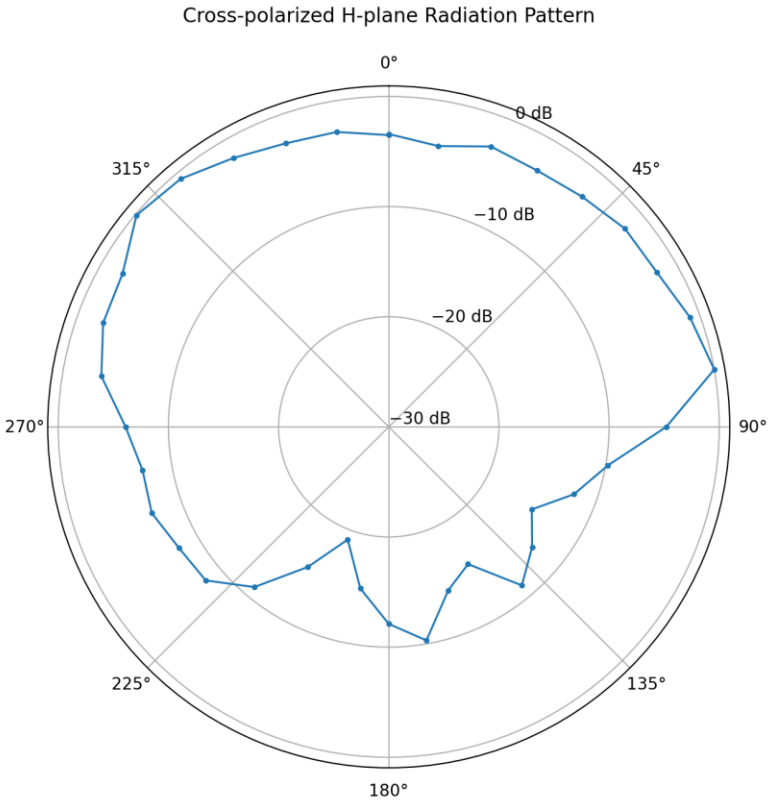
Radiation-pattern observations

Angle (°)	Co-pol H (dBm)	Co-pol E (dBm)	Cross-pol E (dBm)	Cross-pol H (dBm)
0	-43.13	-58.50	-36.73	-51.50
10	-42.81	-61.30	-36.79	-52.20
20	-42.60	-68.70	-37.65	-51.00
30	-43.23	-66.20	-39.09	-51.20
40	-45.01	-64.20	-40.77	-50.80
50	-46.32	-63.60	-41.96	-50.10
60	-47.26	-61.40	-45.16	-50.00
70	-48.71	-61.20	-45.97	-49.00
80	-49.89	-70.10	-50.75	-48.09
90	-57.07	-68.30	-62.16	-52.90
100	-59.63	-62.10	-54.53	-57.90
110	-67.71	-60.50	-50.10	-60.20
120	-53.41	-59.70	-50.90	-63.10
130	-55.92	-60.70	-53.80	-61.10
140	-52.86	-59.30	-52.60	-59.30
150	-52.46	-59.10	-49.77	-63.70
160	-53.22	-59.00	-46.33	-62.30
170	-56.74	-60.50	-45.44	-58.40
180	-54.04	-57.70	-44.71	-60.20
190	-53.09	-56.60	-44.71	-63.20
200	-55.86	-57.70	-48.88	-67.20
210	-59.03	-60.00	-52.38	-63.40
220	-62.23	-68.20	-54.92	-59.10
230	-52.31	-66.40	-56.47	-56.40
240	-50.53	-62.20	-52.67	-56.10
250	-50.21	-61.10	-51.63	-55.20
260	-54.32	-61.20	-52.09	-55.40
270	-52.44	-60.50	-63.35	-54.20
280	-59.48	-75.30	-60.65	-51.60
290	-52.96	-64.90	-46.45	-50.50
300	-52.41	-59.10	-43.13	-50.20
310	-56.42	-56.20	-41.41	-48.20
320	-52.95	-55.90	-40.97	-48.70
330	-48.07	-56.90	-39.74	-49.90
340	-45.89	-57.00	-38.36	-50.70
350	-44.46	-57.10	-37.12	-50.90
360	-44.32	-58.70	-36.72	-51.60









6.Calculations

For beam pattern antennas (horn and dipole antenna)
Length of RF antenna dipole = 4cm
Hence ,

$\frac{\lambda}{4} = 4$. So, $\lambda = 16cm$

$Operating\ Frequency\ = \frac{3 \times 10^8}{16 \times 10^{-2}}$ = 1.087Ghz

Actual operating frequency = 1.42Ghz

$Error\ \ estimation\ \ in\ length(due\ \ to\ cover\ in\ hornantenna) = \lambda\ = \frac{3 \times 10^8}{1.42 \times 10^9}$ = 0.211m = 21.1cm

Error in length = $\frac{21.1 - 16}{21.1}$ = 0.24 .

- VSWR, reflection coefficient and return loss at 300

Vmax = 6 × 500 mV = 3.0 V; Vmin = 4 × 500 mV = 2.0 V

VSWR = $\frac{3.0}{2.0} = 1.50$

$|\Gamma| = \frac{(3.0-2.0)}{(3.0+2.0)}$ = 0.200

RL = $-20\ log_{10}(0.200)$ = 13.98 dB

VSWR, reflection coefficient and return loss at 120

Vmin = 0.8 × 500 mV = 0.40 V; Vmax = 4.5 × 1 V = 4.50 V ; VSWR = 4.50/0.40 = 11.25

$|\Gamma| = \frac{(4.50-0.40)}{(4.50+0.40)}$ = 0.837

RL = $-20 \log_{10}(0.837)$ = 1.55 dB

- E-plane HPBW

Maximum co-polarized E-plane power = -55.90 dBm at approximately 320°. The −3 dB level is -58.90 dBm. Linear interpolation between the measured points gives the two half-power angles as approximately 300.69° and 360.77°.

The half-power point is 3 dB below the maximum.

Therefore,

$$P_{-3dB} = P_{\max} - 3$$

For example:

$P_{-3dB} = -55.9 - 3$ = - 58.9 dBm (at around 300°)

At 290°: $P = -64.9$ dBm and At 300°: $P = -59.1$ dBm
Our required value is: -58.9 dBm
So the crossing is just after 300°.

Linear interpolation expression = $\theta = \theta 1 + \frac{Ptarget-P1}{P2-P1}(\theta 2 - \theta 1)$

Put the values:

$$\theta_L = 300 + \frac{-58.9 - (-59.1)}{-56.2 - (-59.1)}(310 - 300)$$

$= 300 + \frac{0.2}{2.9} \times 10 \approx 300.69°.$

We have: $360° \rightarrow -58.7$ dBm and $10° \rightarrow -61.3$ dBm
Our required value is −58.9 dBm.

$$\theta R = 360 + \frac{-61.3 - (-58.7)}{-58.9 - (-58.7)}(10)$$

$= 360 + \frac{-0.2}{-2.6}(10) \approx 360.77°$.
HPBWE=360.77−300.69 = 60.08

Therefore, θ_E = 60.08° = 1.0485 rad.

- H-plane HPBW

Similarly , Maximum co-polarized H-plane power = -42.60 dBm at approximately 20°. The −3 dB level is -45.60 dBm. Linear interpolation gives the half-power angles as approximately 342.03° and 44.50°. $(360 - 342.03° = 17.97°)$

Therefore, $\theta_H = (17.97° + 44.50°) = 62.48°$ = 1.0904 rad.

- Directivity

$D \approx \frac{4\pi}{\theta_E \theta_H} = \frac{4\pi}{[(1.0601)(1.0904)]} = 10.87$

In decibels, directivity ≈ 10 $\log_{10}$(D) = 10.36 dBi.

6. Result and Conclusion

Main results obtained :

VSWR at 300	**1.50**
Reflection coefficient at 300	**0.200**
Return loss at 300	**13.98 dB**
VSWR at 120	**11.25**
Reflection coefficient at 120	**0.837**
Return loss at 120	**1.55 dB**
E-plane HPBW	**60.08°**
H-plane HPBW	**62.48°**
E-plane HPBW	**1.060 rad**
H-plane HPBW	**1.090 rad**
Estimated directivity	**10.87**
Directivity in dBi	**10.36 dBi**

3.The estimated antenna directivity is D ≈ 10.87, corresponding to about 10.36 dBi.

4. The measured peak received power decreases as the separation between the transmitting and receiving antennas increases.

The antenna was characterized through VSWR/return-loss measurements, received-power measurements at different distances, and angular radiation-pattern measurements. The co-polarized E- and H-plane patterns were used to obtain the −3 dB beamwidths, from which the antenna directivity was estimated. The cross-polarized patterns were also recorded for comparison. Small deviations from an ideal pattern can arise from alignment errors, reflections from nearby objects, instrument uncertainty, and limited angular sampling.

This experiment helped in understanding the practical characterization of an antenna and how its performance can be measured experimentally rather than only from theoretical specifications. By measuring VSWR and return loss, we learned how impedance matching affects the amount of signal reflected back from an antenna. The distance measurement showed how the received power changes with separation between transmitting and receiving antennas. The radiation-pattern measurements helped us understand how an antenna radiates or receives electromagnetic energy in different directions and the difference between co-polarization and cross-polarization.

Overall, the experiment provided a useful practical connection between **electromagnetic theory, antenna parameters, RF instrumentation and actual antenna measurements**, while also showing how antenna characteristics are important in communication, sensing and radio-astronomical applications.